Università degli Studi di Napoli

Federico II

Dipartimento di Ingegneria Elettrica e delle Tecnologie

dell'Informazione

Statistical Learning and Data Analysis – Module A

DATA DESCRIPTION

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A basic concept

- The first step of **Data Processing** is to convert *Data into Information* through Descriptive Statistics.
- **Descriptive Statistics** are methods for organizing and summarizing data such to provide more insights and information about the case study.

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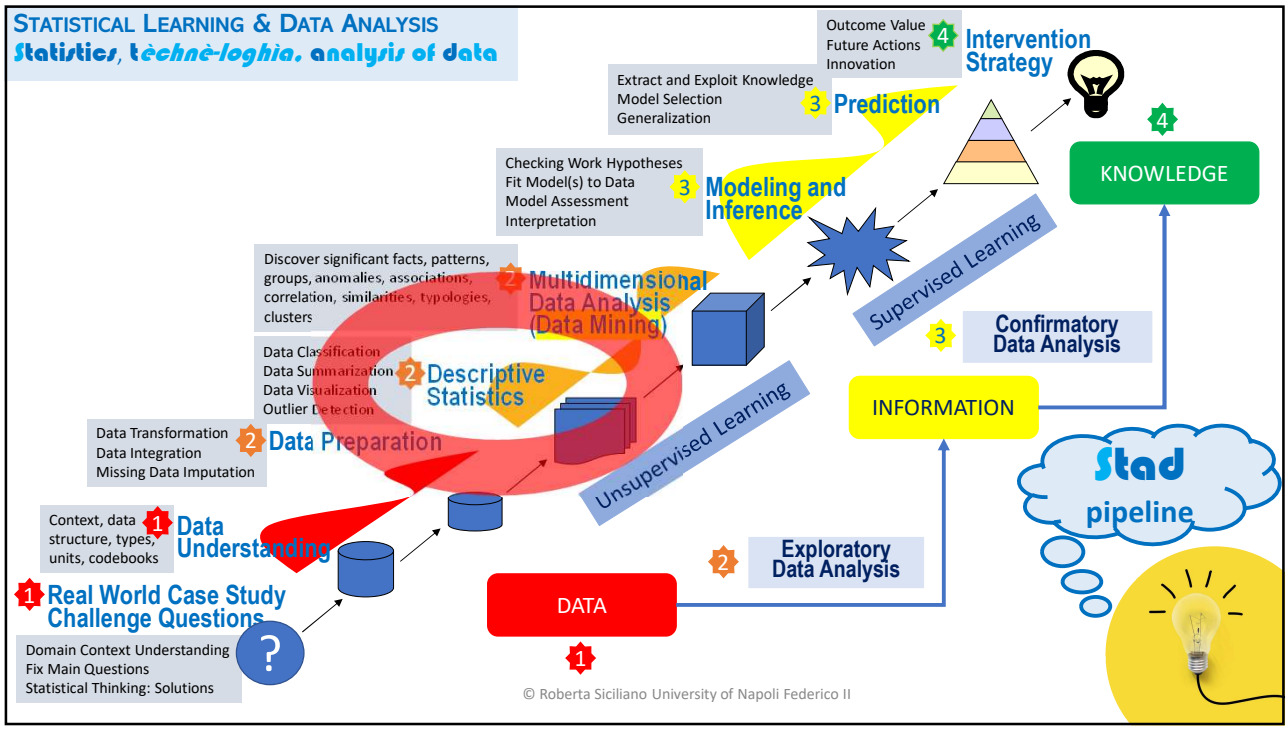

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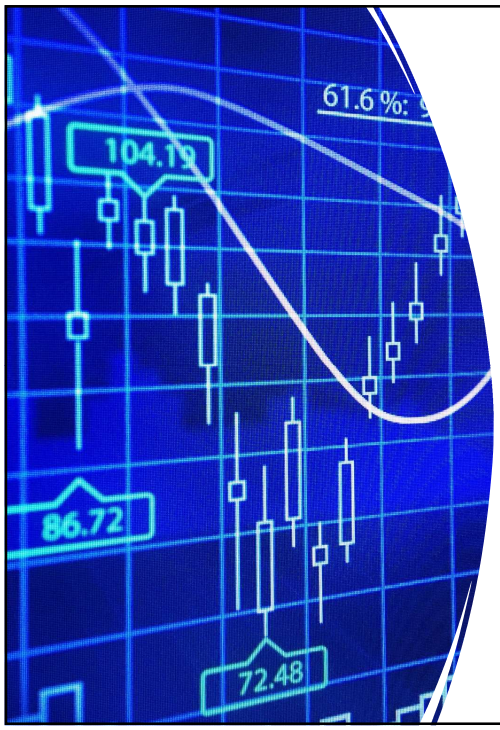

How to describe data

- **Data Classification in Tables** (i.e., frequency distribution table, intensity distribution table)
- **Statistical Indexes** (i.e., mean, variance, standard deviation, asymmetry, kurtosis)
- **Data Visualization** (i.e., bar chart, histogram, box-plot)

Data Visualization will be discussed in a next Lecture!

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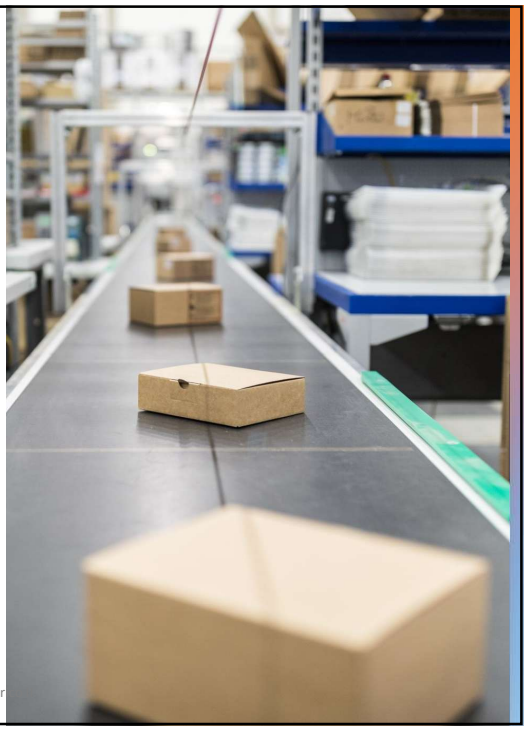
Unitary distribution

Let X be a variable measured on n statistical units with unitary distribution defined as

$x_1, x_2, .., x_l, \dots, x_n$

where x_l is the observed modality of X on the l-th unit.



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Data Classification: Frequency Distribution Table

- **Distinct modalities** are the possible ways the variable can be *"expressed"* and they need to be accompanied by their frequencies yielding to a table of ***frequency distribution of the variable.***
- The **frequency** of a distinct modality is *the number of occurrences of that distinct modality in the raw data.*
- The table of frequency distribution of the variable is a summary of raw data providing a classification of statistical units into distinct groups given by the distinct modalities of the variable.

X	Absolute Frequencies	Relative Frequencies
x_1	n_1	f_1
…	…	…
x_i	n_i	f_i
…	…	…
x_k	n_k	f_k
	n	1

X	Absolute Frequencies	Relative Frequencies
$x_0 - \lvert x_1$	n_1	f_1
…	…	…
$x_{i-1} - \lvert x_i$	n_i	f_i
…	…	…
$x_{k-1} - \lvert x_k$	n_k	f_k
	n	1

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Data Summarization: Statistical Indices

Statistical indices for **data summarization** can be defined to identify:

- the central tendency, or more properly *the most representative modality*;
- the variation or heterogeneity, namely *the attitude of a variable to assume distinct modalities*;
- *the shape of the distribution for numerical variables*.

The most challenging is to visualize the behavior of the unitary variable distribution.

There is no need of data summarization in two special cases:

- If all units have the same value of the numerical variable there is no variation; the central tendency is just the value observed by all units.
- Analogously, if all units belongs to only one category of the qualitative variable that modality is the only representative modality.

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Hierarchy of the variable typology

- The scales of measurements allow to define a **hierarchy of the variable typology** with respect to
  - *the possible operations to be computed on the modalities,*
  - *the information capability,*
  - *the statistical methods that can be applied.*

✓ *Higher is the level in the hierarchy more information can be deduced and more statistical methods can be used.*

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General Rule of Data Processing

- Data processing methods for nominal variables can be performed for all other variables;
- Methods for ordinal variables can be performed also for numerical variables;
- Methods for numerical variables are only designed for numerical variables.

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Measures of Representativeness

Mode

- The modality occurring most often.

Median

- The modality of the (median) unit dividing the ordered distribution into two equal parts.

Mean

- The arithmetic average of numerical values.

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Mode

The **Mode (modality)** *is the most frequent modality of the variable.*

It can be easily determined by the distinct modality associated to the highest frequency in the frequency distribution table.

It is commonly used for nominal variables but can be determined for any type of variable.

In case of continuous variables, it is possible to derive the **Modal Class** of values corresponding to the *highest frequency density*. It is practice to take the semisum of the extreme values of the Modal class to determine a Mode value.

For numerical variables, the Mode is the value that minimizes the following function:

$$\min_c S(c) = \sum_{l=1}^{n} |x_l - c|^0$$

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Median

Let X be an ordinal or numerical variable such that is possible to order the units on the basis of the observed modalities:

$x_{(1)} \leq \cdots \leq x_{(l)} \leq \cdots x_{(n)}$

where the $x_{(l)}$ is the modality of the l -th ordered unit.

The **Median (modality)** *is the observed modality in the middle of the ordered distribution.* The unit in the middle is often called Median Unit.

If there are an *odd number* of units, there is only one unit in the middle, in the position $(n+1)/2$.
If there are an *even number* of units, there are two units in the middle in the positions $n/2$ and $(n/2+1)$ and the two observed modalities could be different.

For numerical variables the Median is the semisum of the central values whereas for ordinal variables there can be two Medians to consider.

For numerical variables, the Median is the value that minimizes the total sum of the absolute differences of the values with respect to a constant c:

$$\min_c S(c) = \sum_{l=1}^{n} |x_l - c|^1$$

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Example

- Suppose the unitary distribution of the numerical variable X is

$x_1 = 5, x_2 = 6, x_3 = 2, x_4 = 8, x_5 = 3.$

- First operation is to order the observed values:

$$x_{(1)} = 2, x_{(2)} = 3, x_{(3)} = 5, x_{(4)} = 6, x_{(5)} = 8\,.$$

- Then the Median is equal to the observed value of the unit in the third position:

$$Me = x_{(3)} = 5.$$

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Position measures for Ordered Scales

- Percentiles
  - The k-th percentile of a set of values divides them such that the $k\%$ of the values fall below and $(100 - k\%)$ of the values fall above.
- Quartiles
  - The first quartile is the 25-th percentile.
  - The second quartile is the Median.
  - The third quartile is the 75-th percentile.

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Quantiles

- The Median can be found in the unitary distribution as

  $Me = x_{(l^*)}$ such that $\frac{l^*}{n} \geq 0.5$ AND $\frac{l^*-1}{n} < 0.5$.
- The Median can be found in the frequency distribution as

  $Me = x_{(i^*)}$ such that $\sum_{i=1}^{i^*} f_i \geq 0.5$ AND $\sum_{i=1}^{i^*-1} f_i < 0.5$.
- The total number of units n can be divided into four parts of equal size so that the (ordered) observed modality at the first quarter is the **First Quartile (modality)** and that at the third quarter is the **Third Quartile (modality)**.
- The total number of units n can be divided into ten parts in order to derive **deciles** of the (ordered) unitary distribution.
- The total number of units n can be divided into hundred parts in order to derive **percentiles** of the (ordered) unitary distribution.
- In general, the **Quantile** of the ordered unitary distribution is the observed (ordered) modality such to leave a quantile proportion of units before:

$Q_\alpha = x_{(l^*)}$ such that $\frac{l^*}{n} \geq \alpha$ AND $\frac{l^*-1}{n} < \alpha$.

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Arithmetic Mean

The Arithmetic Mean, often called Mean, is the most popular index to summarize the central tendency of the numerical variable.

Let X be a numerical variable assuming the observed values $x_1, x_2, .., x_l, \dots, x_n$ on n units.

The **Arithmetic Mean** *is the sum of all values of a numerical variable divided by the total number of units:* $\mu = \frac{1}{n}\sum_{l=1}^{n} x_l$.

The Mean can be understood as the value that substituted to all observed values yields to the same total sum of all values of the unitary distribution **(equidistribution interpretation and property)**. It follows from the identity:

$$\sum_{l=1}^{n} \mu = \sum_{l=1}^{n} x_l \Leftrightarrow n\mu = \sum_{l=1}^{n} x_l$$

The Arithmetic Mean is also the value that minimizes the total sum of squares of all values with respect to a constant c **(barycenter interpretation and property):** $\min_c S(c) = \sum_{l=1}^{n} |x_l - c|^2$

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Chisini's identity

- More in general, according to the Chisini's identity, the Mean is the value M such that

$$f(x_1, x_2, .., x_l, \dots, x_n) = f(M, \dots, M)$$

- Suitable definitions of the function f are:
  - ✓ the sum of all values (Arithmetic Mean M),
  - ✓ the product of all values (Geometric Mean GM),
  - ✓ the sum of the reciprocal values (Harmonic Mean HM),
  - ✓ the squares of the values (Quadratic Mean $M2$).
- It holds that $M2 \geq M \geq GM \geq HM$.

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One more insight about the Mean

- ***Equidistribution interpretation*** holds only for **transferable variables** where in theory an amount of the total sum can be transferred from one unit to another.
- This is not the case for a variable such as the age.
- Then the ***barycenter interpretation*** should be preferred to find *the most representative value (not necessarily coincident to one of the observed values)* located to be equidistant from all other values when using the squared Euclidean distance to compute the proximity between two values.

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Mean in the Frequency Distribution

The Arithmetic Mean can be computed from the frequency distribution of the variable X described as (x_i, n_i) for $i = 1, \dots, k$, where x_i is the i-th distinct modality and n_i its absolute frequency such that $\sum_{i=1}^{k} n_i = n$:

$$\mu = \frac{1}{n}\sum_{i=1}^{k} x_i\, n_i$$

In case of relative frequency distribution, the Mean is equal to $\mu = \sum_{i=1}^{k} x_i\, f_i$ being $f_i = \frac{n_i}{n}$ such that $\sum_{i=1}^{k} f_i = 1$.

For continuous variables the frequency distribution is organized into frequencies associated to the interval values of the variable.

In that case each distinct modality is the center of each interval value (namely, the *semisum* of the two extremes of the interval). The resulting mean is *an approximation* of the Mean computed on the raw data.

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Properties of the Mean

- **Equidistribution property:** $n\mu = \sum_{l=1}^{n} x_l$
- **Barycenter property:** $\min_{c} S(c) = \sum_{l=1}^{n} |x_l - c|^2 \equiv S(\mu)$
- **Invariance to linear transformations:** $Y = a + bX \rightarrow \mu_Y = a + b\mu_X$
- Assume to partition the observed values of the variable Y into k groups on the basis of the k distinct modalities of the variable X. It holds:

$$\mu = \frac{1}{n}\sum_{i=1}^{k} \mu_{Y|X=x_i}\, n_i$$

- where $\mu_{Y|X=x_i}$ is the Mean of the observed values of the variable Y given the i-th group defined by the i -th distinct modality of the variable X, n_i is the number of units in the the i-th group.

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Trimmed Mean

- The Mean is to be preferred for symmetric distribution without extreme values (outliers).
- The **Trimmed Mean** μ_α can be computed cutting off the $\alpha/2\%$ of values in each tail of the distribution providing *a robust central tendency of the variable in presence of outliers.*

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Quality of the Representativeness: *The attitude of a variable to assume "diverse" modalities*

Heterogeneity, Entropy

Interquartile Difference

Range, Variation, Deviance, Standard Deviation, Variation Coefficient

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Heterogeneity

Heterogeneity can be analyzed for any type of variable. It considers only the frequencies of the distribution.

The concept of heterogeneity is fundamental to understand the quality of the Mode to represent the overall distribution: If there are 80 females and 20 males in one distribution and 51 females and 49 males in a second distribution, the Mode "female" is more representative in the first distribution with respect to the second one.

The two complementary concepts of homogeneity and heterogeneity need to be introduced.

The collective of units is **perfectly homogeneous** with respect to the variable X if all units have the same distinct modality whereas it is **perfectly heterogeneous** if the units are equally distributed among the distinct modalities of X.

How good is the representativeness of the Mode is measured by a heterogeneity index: *lower is the degree of heterogeneity better is the quality of representativeness of the Mode.*

Two are the most popular heterogeneity indexes:

Gini Heterogeneity Index: $H = 1 - \sum_{i=1}^{k} f_i^2$ where $0 \leq H \leq \frac{k-1}{k}$. Normalized Gini index [0,1] can be obtained by dividing the index H by its maximum.

Shannon Entropy Index: $H' = -\sum_{i=1}^{k} f_i \log f_i$.

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Range of Values *versus* Interquartile Difference

- **Range of values** considers only the extreme values:
$$R = x_{(max)} - x_{(min)} = x_{(n)} - x_{(1)}$$
- **Interquartile Difference** considers the range of the half part of values in the center of the distribution:
$$IQD = Q_{(0.75)} - Q_{(0.25)}$$

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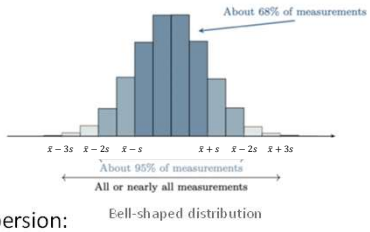
Variance, Standard Deviation, Deviation, Coefficient of Variation

Variance gives insights about the quality of the arithmetic mean to be the barycenter of the distribution showing the amount of dispersion of all values around the barycenter with respect to the number of units:

- $\sigma^2 = \frac{1}{n}\sum_{l=1}^{n}(x_l - \mu)^2 = \frac{1}{n}\sum_{l=1}^{n} x_l^2 - \mu^2$
- $\sigma^2 = \frac{1}{n}\sum_{i=1}^{k}(x_i - \mu)^2 n_i = \frac{1}{n}\sum_{i=1}^{k} x_i^2 n_i - \mu^2$

Standard Deviation is the square root of the variance in order to quantify the above dispersion in the same scale of measure of the variable:

- $\sigma = \sqrt{\sigma^2}$



Deviation is the numerator of the variance to measure the overall amount of dispersion:

- $Dev(X) = n\sigma^2 = \sum_{l=1}^{n}(x_l - \mu)^2 = \sum_{l=1}^{n} x_l^2 - n\mu^2$ (unitary distribution)
- $Dev(X) = n\sigma^2 = \sum_{i=1}^{k}(x_i - \mu)^2 n_i = \sum_{i=1}^{k} x_i^2 n_i - n\mu^2$ (frequency distribution)

For sample data the mean is denoted by $\bar{x}$ and standard deviation is equal to $s_X = \sqrt{\frac{\sum_{l=1}^{n}(x_l - \bar{x})^2}{n-1}}$

Coefficient of Variation is a relative variation index without unit scale: $CV = \frac{\sigma}{|\mu|}$

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Properties of Variance

- **Variance:** $\sigma^2 \geq 0$
- Assume that the variable X is transferable. For a given distribution of values, the total amount is equal to $\sum_{l=1}^{n} x_l = n\mu$. Two special cases are:
  - *Equidistribution:* All units have the same value which is the mean μ.
  - *Maximum concentration:* One unit has the total amount $\sum_{l=1}^{n} x_l = n\mu$ and the remaining $n-1$ have zero. It can be shown that the maximum variance is equal to $\sigma_{max}^2 = (n-1)^2\mu^2$.

➢ A normalized variance index $[0,1]$ is a **concentration index** obtained as $C = \frac{\sigma^2}{\sigma_{max}^2} = \frac{\sigma^2}{(n-1)^2\mu^2} = \frac{CV^2}{(n-1)^2}$ where CV is the Coefficient of Variation.

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Properties of Deviation

- Deviation: $Dev(X) \geq 0$.
- Assume to partition the observed values of the variable Y into k groups on the basis of the k distinct modalities of the variable X. The **Total Deviation** of the variable Y is equal to the sum of Between-Group Deviation and Within-Group Deviation, as it holds:

$$Dev(Y) = Dev(B) + Dev(W)$$

- $Dev(B) = \sum_{i=1}^{k} \left(\mu_{Y|X=x_i} - \mu_Y\right)^2 f_{i+}$ **(Between-Group Deviation)**
- $Dev(W) = \sum_{i=1}^{k} \sum_{j=1}^{h} \left(y_j - \mu_{Y|X=x_i}\right)^2 f_{ij}$ **(Within-Group Deviation)**

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Standardization

Standardization is to transform a numerical variable X into a new variable defined as

$$Z_X = \frac{X - \mu_X}{\sigma_X}$$

such that it holds:

- $\mu_{Z_X} = 0$
- $\sigma^2_{Z_X} = 1$

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Shape of Unitary Distribution

The shape of the distribution of the numerical variable is described by the following characteristics:

- **Symmetry.** When it is graphed, a symmetric distribution can be divided at the center so that each half is a mirror image of the other.
- **Number of peaks.** Distributions can have few or many peaks. Distributions with one clear peak are called **unimodal**, and distributions with two clear peaks are called **bimodal**. When a symmetric distribution has a single peak at the center, it is referred to as ***bell-shaped***.
- **Skewness.** When they are displayed graphically, some distributions have many more observations on one side of the graph than the other. One tail is much longer than the other and observations are more concentrated on either low values or high values. Distributions with fewer observations on the right (toward higher values) are said to be ***skewed to the right***; and distributions with fewer observations on the left (toward lower values) are said to be ***skewed to the left***.
- **Kurtosis.** It is all about the tails of the distribution – not the peaked-ness or flatness. It is used to describe the extreme values in one versus the other tail.

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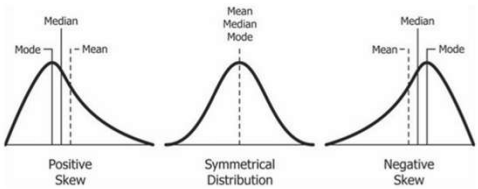
Skewness

It measures the asymmetry of data:

- Positive or right skewed: *Longer right tail*
- Negative or left skewed: *Longer left tail*

Different measures of skewness:

- *Pearson's coefficient of skewness* (gap mean/mode/median)
- *Bowley's coefficient of skewness* (use quartiles)
- *Kelly's coefficient of skewness* (use only the central 50% units)



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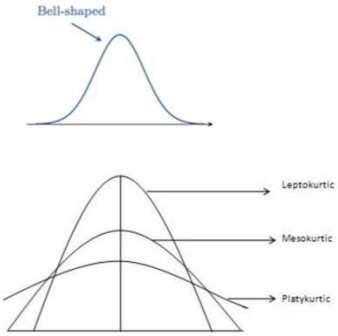
Kurtosis

- The degree of sharpness of a distribution is measured by kurtosis. Traditionally the value of this coefficient is compared to a value of 0.0, which is the coefficient of kurtosis for a normal (Gaussian) distribution, i.e., the bell-shaped curve.
- **High kurtosis** in a data set is an indicator that data has heavy tails or outliers. If there is a high kurtosis, then, we need to investigate why do we have so many outliers. It indicates a lot of things, maybe wrong data entry or other things. Investigate!
- **Low kurtosis** in a data set is an indicator that data has light tails or lack of outliers. If we get low kurtosis (too good to be true), then also we need to investigate and trim the dataset of unwanted results.



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